

Computational Investigations of the Riemann Hypothesis: Parameterized Operator Search, Spectral Geometry, and Formal Verification

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Abstract

We present a computational investigation of the Riemann Hypothesis from multiple angles. Our main contributions are: (1) the first systematic computational search over the parameterized operator family $H = U(x)\hat{p} + V(x)/\hat{p}$, finding that the sub-linear exponent $\alpha \approx 0.9$ in $U(x) = ax^\alpha$ yields WKB counting functions that match individual zeta zeros $2.6\times$ better than the Berry–Keating value $\alpha = 1$; (2) a precise measurement of the scale-dependent effective spectral dimension $d_{\text{eff}} \approx 2.36\text{--}3.52$ from the heat trace of zeta zeros, constraining the Hilbert–Pólya operator to non-commutative or fractal geometries; (3) an independent numerical verification of the Connes–Consani–Moscovici (2025) zeta spectral triple construction; (4) a Lean 4 formalization of Robin’s and Lagarias’ inequalities as equivalences with the Riemann Hypothesis, compiling against Mathlib; (5) a high-precision computation of the Nyman–Beurling–Baéz–Duarte distance d_N through $N = 5000$ (dense to $N = 2000$, sparse thereafter) using a fully analytic Gram-matrix formula, reaching $d_{5000} = 0.0723$ with all matrices full rank; and identifying a previously reported apparent decay rate $\alpha \approx 0.308$ as a numerical artifact of adaptive quadrature on oscillatory integrands, with the corrected rate $\alpha \approx 0.071$ ($N \geq 1000$) still not stabilized; and (6) an unconditional analytic proof that the Li coefficients satisfy $\lambda_n > 0$ for all $n \leq 85$ (Theorem 2), based on a boundary characterization of the functional pair contribution $F_n(\sigma, \gamma)$; an extension certifying $\lambda_n > 0$ for all $n \leq 2824$ outside a set S of 1126 exceptional values using the classical zero-free region of Mossinghoff–Trudgian–Yang [21] and a periodic failure-window analysis covering thirty-one failure windows, with a complete computational scan to $n = 10^7$ finding exactly 145 certifiable runs containing 13,881 certifiable right endpoints (windows 32–537 all certifiable in run 1, followed by a quasi-periodic sequence of certifiable runs and barriers covering $n \leq 9,834,937$ in aggregate) (Corollary 5); and a multi-zero extension certifying $\lambda_n > 0$ for all $n \leq 7.1 \times 10^6$ using the Odlyzko 2M verified-zero dataset (first 2,001,052 zeros, reaching height $\gamma \approx 1.13 \times 10^6$), and further to all $n \leq 4.83 \times 10^{12}$ using the Platt–Trudgian verification of the first 3×10^{12} zeros [2] (Theorem 9); and (7) an unconditional numerical certificate via direct Cauchy DFT evaluation of $\xi(s)$ on $|s| = 5$, establishing $\lambda_n > 0$ for all $n \leq 10,000$ with total error $< 3 \times 10^{-19}$ and no assumption on zero locations (Section 4).

1 Introduction

The Riemann Hypothesis (RH), conjectured in 1859 [1], asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ have real part $\frac{1}{2}$. Despite numerical verification for the first 10^{13} zeros [2] and its equivalence to numerous statements across number theory, analysis, and mathematical physics, a proof remains elusive.

The Hilbert–Pólya approach seeks a self-adjoint operator H whose eigenvalues are the imaginary parts γ_n of the non-trivial zeros. Self-adjointness forces real eigenvalues, which would place all

zeros on the critical line. The discovery by Montgomery [3] that the pair correlation of zeta zeros matches the Gaussian Unitary Ensemble (GUE) of random matrix theory, confirmed numerically by Odlyzko [4], suggests this operator resembles the Hamiltonian of a quantum chaotic system. A complementary function-theoretic reformulation, due to Nyman [17] and extended by Baéz-Duarte [16], characterizes RH by the vanishing of a distance d_N in a Hilbert space of fractional-part functions.

In this paper, we approach RH computationally through six independent directions, each producing concrete artifacts: numerical data, visualizations, operator candidates, machine-verified formal statements, and high-precision analytic Gram-matrix computations.

2 Data and Statistical Verification

2.1 Zero datasets

We work with three datasets of non-trivial zeta zeros:

- 100,000 zeros from LMFDB at 31 decimal places (high-precision reference),
- 100,000 zeros from Odlyzko’s tables at 9 decimal places,
- 2,001,052 zeros from Odlyzko’s tables at 9 decimal places.

Our pipeline was verified against independent computation via `mpmath.zetazero()` to 10^{-31} precision for the first 100 zeros.

2.2 GUE statistics

We computed normalized spacings using the smooth unfolding $\hat{\gamma}_n = \bar{N}(\gamma_n)$ where $\bar{N}(E) = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + \frac{7}{8}$.

The spacing variance converges toward the GUE Wigner surmise prediction of 0.178 (not 0.286, which is the GOE value—a distinction frequently confused in computational references). At 100,000 zeros the empirical variance is 0.161; the 2,000,000-zero dataset shows this increasing with height on the critical line from 0.162 at height $\sim 75,000$ to 0.168 at height $\sim 1,080,000$, consistent with asymptotic GUE behavior.

We also computed the exact GUE spacing distribution via the Fredholm determinant of the sine kernel, finding mean absolute deviation 0.017 from the empirical distribution—comparable to the Wigner surmise approximation (0.017), confirming that at this sample size the two are nearly indistinguishable.

The pair correlation function matches the Montgomery–Odlyzko law $R_2(x) = 1 - \left(\frac{\sin \pi x}{\pi x}\right)^2$ to mean deviation 0.023 after correcting a normalization error in the initial computation.

2.3 Li coefficients

We computed partial sums of the first 50,000 Li coefficients $\lambda_n^{(50k)} = \sum_{j=1}^{50000} F_n(\sigma_j, \gamma_j)$ using the Odlyzko zero data (assuming RH). All are positive, consistent with RH (minimum $\lambda_1 \approx 0.02306$). Key partial-sum values: $\lambda_{100}^{(50k)} \approx 118.2$, $\lambda_{1000}^{(50k)} \approx 2287.6$, $\lambda_{10000}^{(50k)} \approx 30897$, $\lambda_{50000}^{(50k)} > 0$. These are partial sums conditional on RH; the full (unconditional) values—which include contributions from all zeros—are given precisely in Section 4 (e.g. $\lambda_{10000} = 34736.5797322416$). The partial-sum truncation error grows predictably with n , consistent with the tail-sum bound $\Delta \lambda_n^{(50k)} \approx n^2(1 + \log \gamma_{50k})/(2\pi\gamma_{50k})$ where $\gamma_{50k} \approx 40434$:

n	$\lambda_n^{(50k)}$	λ_n (uncond.)	tail error	error%
10	2.2755	2.2793	0.004	0.17%
100	118.2	118.604	0.38	0.32%
500	982.3	991.900	9.6	0.97%
1000	2287.6	2326.053	38.5	1.65%
2000	5198.0	5351.760	153.8	2.87%
5000	14678	15639.094	961	6.1%
10000	30897	34736.580	3840	11.1%

The ratio of the unconditional λ_n to the Keiper–Li asymptotic $\lambda_n \sim \frac{n}{2}(\log n + \gamma_E - 1 - \log 2\pi)$ converges to 0.997 at $n = 200$ and approaches 1 monotonically.

3 Unconditional Positivity of λ_n : Pair Positivity

3.1 Functional pair structure

For each non-trivial zero $\rho = \sigma + i\gamma$ the functional equation forces $1 - \bar{\rho}$ to also be a zero. Define the pair contribution

$$F_n(\sigma, \gamma) = C_n(\sigma, \gamma) + C_n(1 - \sigma, \gamma), \quad C_n(\sigma, \gamma) = 2 \operatorname{Re} \left[1 - \left(\frac{\rho - 1}{\rho} \right)^n \right].$$

Then $\lambda_n = \sum_{\text{pairs}} F_n(\sigma_j, \gamma_j)$. At $\sigma = 0$,

$$F_n(0, \gamma) = 4(1 - \cosh(n \log r) \cos(n \arctan(1/\gamma))), \quad r = \sqrt{1 + \gamma^{-2}}.$$

The key algebraic identity $|\alpha(\sigma)| \cdot |\alpha(1 - \sigma)| = 1$, where $\alpha(\sigma) = (\rho - 1)/\rho$, holds for *all* $\sigma \in (0, 1)$. This means $\sigma = 0$ gives $|\alpha| = r > 1$ and $\sigma = 1$ gives $|\alpha| = 1/r < 1$, with $\sigma = 1/2$ giving $|\alpha| = 1$ (the on-line case).

Lemma 1 (Phase formula). *For $\rho = \sigma + i\gamma$ with $\sigma \in (0, 1)$ and $\gamma > 0$,*

$$F_n(\sigma, \gamma) = 4 \left(1 - \cosh(n \log r(\sigma, \gamma)) \cos(n \varphi(\sigma, \gamma)) \right),$$

where $r(\sigma, \gamma)^2 = ((1 - \sigma)^2 + \gamma^2)/(\sigma^2 + \gamma^2)$ and

$$\varphi(\sigma, \gamma) = \arctan \left(\frac{\gamma}{\gamma^2 - \sigma(1 - \sigma)} \right).$$

The phase $\varphi(\sigma, \gamma)$ is maximized at $\sigma = \frac{1}{2}$, with $\varphi(0, \gamma) = \arctan(1/\gamma)$ and $\varphi(\frac{1}{2}, \gamma) = \arctan(4\gamma/(4\gamma^2 - 1))$. The phase increment satisfies $\varphi(\frac{1}{2}, \gamma) - \varphi(0, \gamma) \approx 1/(4\gamma^3)$ for large γ .

Proof. The phases of $\alpha_1 = (\rho - 1)/\rho$ and $\alpha_2 = (\rho_c - 1)/\rho_c$ (with $\rho_c = (1 - \sigma) + i\gamma$) are equal: $\arg(\alpha_1) = \arg(\alpha_2) = \pi - \arctan(\gamma/(1 - \sigma)) - \arctan(\gamma/\sigma)$. By the addition formula for $\arctan$, $\pi - \arctan(\gamma/(1 - \sigma)) - \arctan(\gamma/\sigma) = \arctan(\sigma/\gamma) + \arctan((1 - \sigma)/\gamma) = \arctan(\gamma/(\gamma^2 - \sigma(1 - \sigma)))$, giving $\varphi(\sigma, \gamma)$ as stated. Since $\sigma(1 - \sigma)$ is maximized at $\sigma = \frac{1}{2}$, $\gamma^2 - \sigma(1 - \sigma)$ is minimized there, making φ maximal at $\sigma = \frac{1}{2}$. $\square$

3.2 Theorem: pair positivity for $n \leq 85$

Theorem 2 (Pair Positivity — Unconditional). *For all $n = 1, \dots, 85$, all $\gamma \geq \gamma_1 = 14.1347\dots$, and all $\sigma \in (0, 1)$:*

$$F_n(\sigma, \gamma) > 0.$$

Consequently, $\lambda_n > 0$ for $n = 1, \dots, 85$ unconditionally (no assumption on RH).

Proof sketch. The tight case is $\sigma \rightarrow 0$, where we must show $\cosh(x)\cos(y) < 1$ with $x = n \log r$, $y = n \arctan(1/\gamma)$.

Step 1 (Boundary characterization). Set $y^* = \arctan(\sinh(x))$. Then $\cosh(x)\cos(y^*) = \cosh(x) \cdot \operatorname{sech}(x) = 1$, so $\cosh(x)\cos(y) < 1$ iff $y \in (y^*, 2\pi - y^*)$, equivalently $y + y^* < 2\pi$.

Step 2 (Lower bound $y > y^*$). (a) $\arctan(t) > \frac{1}{2} \log(1 + t^2)$ for $t \in (0, 1)$ gives $y > x$; (b) $\arctan(\sinh(x)) < x$ for $x > 0$ gives $y^* < x$. Together: $y > x > y^*$.

Step 3 (Upper bound $y + y^* < 2\pi$). Both $\arctan(1/\gamma)$ and $\log r$ are decreasing in γ , so the left-hand side is maximized at $\gamma = \gamma_1$. Computation at $(n = 85, \gamma_1)$ with 50-digit `mpmath` precision gives

$$y + y^* = 6.2142\dots < 2\pi = 6.2832\dots, \quad \text{gap} = 0.0690 > 0.$$

Threshold sharpness: at $n = 86$, $y + y^* = 6.2872 > 2\pi$ (condition fails).

Extension to all $\sigma \in (0, 1)$: A dense numerical scan over 200 values of σ , 700 values of $\gamma \in [\gamma_1, 1000]$, and all $n = 1, \dots, 85$ confirms $F_n(\sigma, \gamma) > 0$ everywhere. For $\gamma > 1000$ and $n \leq 85$, the boundary condition $n \arctan(1/\gamma) + \arctan(\sinh(n \log r)) \leq 85/1000 + O(1/\gamma^2) \ll 2\pi$ holds trivially, so $F_n(0, \gamma) > 0$; the monotonicity argument then gives $F_n(\sigma, \gamma) > 0$. (Equivalently, for large γ : $F_n(0, \gamma) \approx 2n^2/\gamma^2 \cdot (1 - 1/(4\gamma^2)) > n^2/\gamma^2 > 0$.) A Lipschitz argument certifies the gaps between grid points. $\square$

Remark 3. *The threshold $n = 85$ is exact: it equals $\lfloor (2\pi - y^*)/\arctan(1/\gamma_1) \rfloor$ where $y^* \approx 0.211$ at γ_1 . This threshold is determined entirely by γ_1 .*

Remark 4 (Extension to $n = 86$ via the classical zero-free region). *The failure at $n = 86$ is a near-miss: $F_{86}(0, \gamma_1) = -0.00348$, barely negative. A refined analysis shows $F_{86}(\sigma, \gamma_1) > 0$ for all $\sigma > \sigma_c \approx 0.0104$. The classical zero-free region of Mossinghoff, Trudgian, and Yang [21] certifies $\zeta(\sigma + it) \neq 0$ for $\sigma \geq 1 - 1/(5.558691 \log |t|)$, $|t| \geq 2$. By the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, non-trivial zeros are symmetric under $\rho \leftrightarrow 1 - \rho$, so the zero-free region near $\sigma = 1$ implies a matching zero-free strip near $\sigma = 0$:*

$$\delta := \frac{1}{5.558691 \log \gamma_1} = \frac{1}{5.558691 \times 2.6486} \approx 0.0679.$$

Since $\delta = 0.0679 \gg \sigma_c = 0.0104$ (a $6.5\times$ margin), all non-trivial zeros near height γ_1 satisfy $\operatorname{Re}(\rho) > 0.0679 > \sigma_c$, giving $F_{86}(\sigma, \gamma_1) > 0$. For all other zeros $\gamma \geq \gamma^$ (where $\gamma^* \approx 14.14359$ is the unique crossing where $F_{86}(0, \cdot)$ changes sign (computed to 50 digits), with no actual zero in (γ_1, γ_2)), $F_{86}(0, \gamma) > 0$ follows from the boundary condition. Therefore $\lambda_{86} > 0$ unconditionally; see Corollary 5 below. By contrast, $n = 87$ requires $\sigma_c \approx 0.19$, far beyond any known zero-free strip.*

Corollary 5 (Unconditional positivity for all $n \leq 2824$ outside thirty-one window interiors). *Let*

$$\begin{aligned}
S = & \{87, \dots, 91\} \cup \{173, \dots, 183\} \cup \{260, \dots, 274\} \cup \{347, \dots, 365\} \cup \{434, \dots, 456\} \\
& \cup \{521, \dots, 547\} \cup \{608, \dots, 637\} \cup \{696, \dots, 727\} \\
& \cup \{784, \dots, 817\} \cup \{872, \dots, 907\} \cup \{960, \dots, 997\} \\
& \cup \{1048, \dots, 1086\} \cup \{1137, \dots, 1176\} \cup \{1225, \dots, 1265\} \\
& \cup \{1314, \dots, 1354\} \cup \{1403, \dots, 1443\} \\
& \cup \{1491, \dots, 1533\} \cup \{1580, \dots, 1622\} \cup \{1669, \dots, 1711\} \\
& \cup \{1758, \dots, 1800\} \cup \{1847, \dots, 1889\} \cup \{1936, \dots, 1978\} \\
& \cup \{2024, \dots, 2067\} \cup \{2113, \dots, 2156\} \cup \{2202, \dots, 2245\} \\
& \cup \{2291, \dots, 2334\} \cup \{2380, \dots, 2423\} \cup \{2469, \dots, 2512\} \\
& \cup \{2558, \dots, 2601\} \cup \{2647, \dots, 2690\} \\
& \cup \{2736, \dots, 2779\}
\end{aligned}$$

(1126 values; the interiors of thirty-one failure windows). Then $\lambda_n > 0$ for all $n \leq 2824$ with $n \notin S$.

Proof. We partition the certified range into two types.

Type I (safe zones): For n in the safe zones $[1, 85]$, $[93, 172]$, $[185, 258]$, $[276, 344]$, $[368, 431]$, $[459, 519]$, $[549, 606]$, $[640, 694]$, $[730, 782]$, $[820, 870]$, $[909, 958]$, $[999, 1047]$, $[1088, 1135]$, $[1178, 1224]$, $[1267, 1313]$, $[1356, 1401]$, $[1445, 1490]$, $[1534, 1579]$, $[1624, 1668]$, $[1713, 1757]$, $[1802, 1846]$, $[1891, 1935]$, $[1980, 2023]$, $[2069, 2112]$, $[2158, 2201]$, $[2247, 2290]$, $[2336, 2379]$, $[2425, 2468]$, $[2514, 2557]$, $[2603, 2646]$, $[2691, 2735]$, $[2780, 2824]$, the worst-case value $F_n(0, \gamma_1) > 0$ (verified to 50-digit precision for each n ; the narrowest margins are $+0.0022$ at $n = 172$ and $+0.0124$ at $n = 276$). For $n \leq 85$ this is exactly Theorem 2 (which includes a dense σ -scan over 200 values of $\sigma \in [0, 1]$ and 700 values of $\gamma \in [\gamma_1, 1000]$). For $n > 85$ in the safe zones, the same methodology applies: since $F_n(0, \gamma_1) > 0$ and both $\arctan(1/\gamma)$ and $\log \sqrt{1 + \gamma^{-2}}$ are strictly decreasing in γ , we have $F_n(0, \gamma) > 0$ for all $\gamma \geq \gamma_1$. A dense scan over 200 values of $\sigma \in [0, 1]$ at each such n then confirms $F_n(\sigma, \gamma) > 0$ uniformly (Lipschitz argument certifies gaps; for $\gamma > 1000$ the boundary condition holds trivially giving $F_n(0, \gamma) > n^2/\gamma^2 > 0$). Hence $F_n(\sigma, \gamma) > 0$ for all $\sigma \in (0, 1)$ and $\gamma \geq \gamma_1$.

Type II (window endpoints): For n in

$$\begin{aligned}
E := & \{86, 92, 184, 259, 275, 345, 346, 366, 367, \\
& 432, 433, 457, 458, 520, 548, \\
& 607, 638, 639, 695, 728, 729, \\
& 783, 818, 819, 871, 908, \\
& 959, 998, 1087, 1136, 1177, \\
& 1266, 1355, 1402, 1444, \\
& 1623, 1712, 1801, 1890, 1979, \\
& 2068, 2157, 2246, 2335, 2424, 2513, 2602\},
\end{aligned}$$

the pair contribution $F_n(0, \gamma_1) < 0$, but only slightly:

n	$F_n(0, \gamma_1)$	$\sigma_c(n)$	margin $\delta - \sigma_c$
86	-0.00348	0.01044	+0.0575
92	-0.01162	0.02694	+0.0410
184	-0.02703	0.01613	+0.0518
259	-0.13178	0.04969	+0.0182
275	-0.18281	0.05303	+0.0149
345	-0.02209	0.00511	+0.0628
346	-0.29223	0.06678	+0.0011
366	-0.36038	0.06288	+0.0050
367	-0.08815	0.01514	+0.0528
432	-0.05878	0.00969	+0.0582
433	-0.42114	0.06698	+0.0009
457	-0.48993	0.05839	+0.0095
458	-0.11755	0.01404	+0.0539
520	-0.43981	0.05407	+0.0138
548	-0.47579	0.04235	+0.0256
607	-0.24386	0.02566	+0.0423
638	-0.82956	0.05748	+0.0104
639	-0.18986	0.01356	+0.0544
695	-0.47927	0.04249	+0.0254
728	-1.12470	0.06279	+0.0051
729	-0.30699	0.01788	+0.0500
783	-0.60649	0.04781	+0.0201
818	-1.28654	0.05960	+0.0083
819	-0.24241	0.01186	+0.0561
871	-0.54215	0.04079	+0.0271
908	-1.21227	0.04783	+0.0201
959	-0.17315	0.01421	+0.0537
998	-0.76110	0.02619	+0.0417
1087	-1.84520	0.05198	+0.0159
1136	-0.26977	0.02485	+0.0431
1177	-0.55682	0.01404	+0.0539
1266	-1.54922	0.03220	+0.0357
1355	-2.75608	0.04722	+0.0207
1402	-0.18550	0.05628	+0.0116
1444	-4.22207	0.05956	+0.0084
1623	-0.08611	0.00088	+0.0670
1712	-0.68725	0.00568	+0.0622
1801	-1.33487	0.00889	+0.0590
1890	-2.01478	0.01077	+0.0572
1979	-2.70290	0.01153	+0.0564
2068	-3.36119	0.01138	+0.0565
2157	-3.93216	0.01053	+0.0574
2246	-4.33157	0.00914	+0.0588
2335	-4.43872	0.00735	+0.0606
2424	-4.08341	0.00529	+0.0626
2513	-3.02892	0.00307	+0.0649
2602	-0.94931	0.00075	+0.0672

Let $\gamma^*(n)$ be the unique $\gamma > \gamma_1$ with $F_n(0, \gamma^*(n)) = 0$. One verifies $\gamma^*(n) < \gamma_2 = 21.022$ for all $n \in E$ (computed to 50 digits; all values lie in $[14.136, 15.101]$, the minimum being $\gamma^*(1136) \approx 14.136$ and the maximum $\gamma^*(92) \approx 15.101$). Hence there are no non-trivial zeros with $|\text{Im}(\rho)| \in (\gamma_1, \gamma^*(n))$, and for $|\text{Im}(\rho)| > \gamma^*(n)$ the Type I argument applies. For the first zero ρ_1 , the classical zero-free region [21] gives

$$\text{Re}(\rho_1) \geq \delta := \frac{1}{5.558691 \log \gamma_1} = 0.0679.$$

Since $\delta > \sigma_c(n)$ for all $n \in E$ (the tightest margin is $+0.0009$ at $n = 433$; all 47 entries verified), we get $F_n(\text{Re}(\rho_1), \gamma_1) > 0$: the function $\sigma \mapsto F_n(\sigma, \gamma_1)$ satisfies $F_n(0, \gamma_1) < 0$ (barrier window) and $F_n(\frac{1}{2}, \gamma_1) \geq 0$ (since $|\alpha(\frac{1}{2}, \gamma_1)| = 1$), so a sign change occurs at $\sigma_c(n)$; the dense scan confirms $F_n(\sigma, \gamma_1) > 0$ for all $\sigma \in (\sigma_c(n), \frac{1}{2}]$. Since $\text{Re}(\rho_1) \geq \delta > \sigma_c(n)$, we conclude $F_n(\text{Re}(\rho_1), \gamma_1) > 0$.

In both types every zero term is positive, so $\lambda_n > 0$. $\square$

Proposition 6 (Exact closed form for $F_n(0, \gamma_1)$). *Let $\theta = \arctan(1/\gamma_1)$, $r = \sqrt{1 + \gamma_1^{-2}}$, and $T = 2\pi/\theta$. Then*

$$F_n(0, \gamma_1) = 4(1 - \cos(n\theta) \cosh(n \log r)).$$

Consequently, n lies in a failure window if and only if $\cos(n\theta) \cosh(n \log r) > 1$.

Proof. At $\sigma = 0$: $\rho = i\gamma_1$ and $\rho_c = 1 + i\gamma_1$. Direct computation gives

$$\alpha_1 = \frac{\rho - 1}{\rho} = 1 + \frac{i}{\gamma_1} = re^{i\theta}, \quad \alpha_2 = \frac{\rho_c - 1}{\rho_c} = \frac{i\gamma_1}{1 + i\gamma_1} = \frac{1}{r}e^{-i\theta},$$

so $|\alpha_1| = r > 1$, $|\alpha_2| = 1/r < 1$, and $\arg(\alpha_1) = \theta$, $\arg(\alpha_2) = -\theta$. Therefore

$$\begin{aligned} F_n(0, \gamma_1) &= 4 - 2\text{Re}(\alpha_1^n) - 2\text{Re}(\alpha_2^n) \\ &= 4 - 2r^n \cos(n\theta) - 2r^{-n} \cos(n\theta) \\ &= 4 - 4\cos(n\theta) \cosh(n \log r). \end{aligned} \quad \square$$

$\square$

Corollary 7 (Window width saturation). *The k -th failure window W_k is centered near $n = kT$ with width converging to $T/2$ as $k \rightarrow \infty$. Specifically, since $r^n \rightarrow \infty$ and $r^{-n} \rightarrow 0$, the failure condition $\cos(n\theta) \cosh(n \log r) > 1$ asymptotically reduces to $\cos(n\theta) > 0$, which holds on an interval of length $T/2$ centered at each kT . The limiting uncertified density (using only γ_1) is $(T/2 - 2)/T \approx 47\%$, not 0; the multi-zero extension (Theorem 9) is required to handle the remaining values.*

Remark 8 (Failure window structure and uncertified values). *The pair contribution $F_n(0, \gamma_1)$ changes sign periodically with period $T = 2\pi/\arctan(1/\gamma_1) \approx 88.9$. The failure windows (where $F_n(0, \gamma_1) < 0$) and interstitial safe zones are:*

k	window W_k	interior S_k	safe zone after W_k	min of F_n
1	{86, ..., 92}	{87, ..., 91}	{93, ..., 172}	-0.099 ($n = 89$)
2	{173, ..., 184}	{173, ..., 183}	{185, ..., 258}	-0.401 ($n = 178$)
3	{259, ..., 275}	{260, ..., 274}	{276, ..., 344}	-0.922 ($n = 267$)
4	{345, ..., 367}	{347, ..., 365}	{368, ..., 431}	-1.686 ($n = 356$)
5	{432, ..., 458}	{434, ..., 456}	{459, ..., 519}	-2.732 ($n = 445$)
6	{520, ..., 548}	{521, ..., 547}	{549, ..., 606}	-4.111 ($n = 534$)
7	{607, ..., 639}	{608, ..., 637}	{640, ..., 694}	-5.893 ($n = 623$)
8	{695, ..., 729}	{696, ..., 727}	{730, ..., 782}	-8.164 ($n = 712$)
9	{783, ..., 819}	{784, ..., 817}	{820, ..., 870}	-11.038 ($n = 801$)
10	{871, ..., 908}	{872, ..., 907}	{909, ..., 958}	-14.657 ($n = 890$)
11	{959, ..., 998}	{960, ..., 997}	{999, ..., 1047}	-19.199 ($n = 979$)
12	{1048, ..., 1087}	{1048, ..., 1086}	{1088, ..., 1135}	-24.890 ($n = 1068$)
13	{1136, ..., 1177}	{1137, ..., 1176}	{1178, ..., 1224}	-32.012 ($n = 1157$)
14	{1225, ..., 1266}	{1225, ..., 1265}	{1267, ..., 1313}	-40.916 ($n = 1246$)
15	{1314, ..., 1355}	{1314, ..., 1354}	{1356, ..., 1401}	-52.045 ($n = 1335$)
16	{1402, ..., 1444}	{1403, ..., 1443}	{1445, ..., 1490}	-65.947 ($n = 1424$)
17	{1491, ..., 1533}	{1491, ..., 1533}	{1534, ..., 1579}	-83.312 ($n = 1513$)
18	{1580, ..., 1623}	{1580, ..., 1622}	{1624, ..., 1668}	-104.998 ($n = 1602$)
19	{1669, ..., 1712}	{1669, ..., 1711}	{1713, ..., 1757}	-132.078 ($n = 1691$)
20	{1758, ..., 1801}	{1758, ..., 1800}	{1802, ..., 1846}	-165.893 ($n = 1780$)
21	{1847, ..., 1890}	{1847, ..., 1889}	{1891, ..., 1935}	-208.114 ($n = 1869$)
22	{1936, ..., 1979}	{1936, ..., 1978}	{1980, ..., 2023}	-260.831 ($n = 1958$)
23	{2024, ..., 2068}	{2024, ..., 2067}	{2069, ..., 2112}	-326.650 ($n = 2047$)
24	{2113, ..., 2157}	{2113, ..., 2156}	{2158, ..., 2201}	-408.827 ($n = 2136$)
25	{2202, ..., 2246}	{2202, ..., 2245}	{2247, ..., 2290}	-511.486 ($n = 2224$)
26	{2291, ..., 2335}	{2291, ..., 2334}	{2336, ..., 2379}	-639.726 ($n = 2313$)
27	{2380, ..., 2424}	{2380, ..., 2423}	{2425, ..., 2468}	-799.864 ($n = 2402$)
28	{2469, ..., 2513}	{2469, ..., 2512}	{2514, ..., 2557}	-999.832 ($n = 2491$)
29	{2558, ..., 2602}	{2558, ..., 2601}	{2603, ..., 2646}	-1249.534 ($n = 2580$)
30	{2647, ..., 2690}	{2647, ..., 2690}	{2691, ..., 2735}	-1561.338 ($n = 2669$)
31	{2736, ..., 2779}	{2736, ..., 2779}	{2780, ..., 2824}	-1950.683 ($n = 2758$)

Window interiors S_k are uncertified: their $\sigma_c(n) > \delta = 0.0679$. The certified boundary values (those with $\sigma_c(n) < \delta$) are $n \in E$ as listed in the proof; for most windows the set E contains at least the right endpoint of W_k (when the exit from the window is gradual enough), and also the left endpoint when the entry is similarly gentle. Doubly uncertified windows ($k = 17, 30, 31$) contribute no elements to E ; their entire interiors lie in S . Windows 2, 12, 14, and 15 are anomalous: their left endpoints ($n = 173, 1048, 1225, 1314$) have $\sigma_c > \delta$ (entering too steeply), so those values fall into S rather than E . Windows 17 ($n = 1491$ – 1533), 30 ($n = 2647$ – 2690), and 31 ($n = 2736$ – 2779) are doubly uncertified: neither endpoint is certifiable (window 17: $\sigma_c(1533) = 0.070 > \delta$; window 30: $\sigma_c(2690) = 0.074 > \delta$; window 31: $\sigma_c(2779) = 0.070 > \delta$), so their entire interiors fall into S . For windows 18–29, only the right endpoint is certifiable (with σ_c as small as 0.00075 at $n = 2602$, window 29), and similarly for each safe zone between consecutive windows. The classical certification thus extends to $n = 2824$ (end of safe zone 31). Beyond $n = 2824$, a computational scan (50-digit bisection) confirms that windows 32–537 all have certifiable right endpoints (with σ_c ranging between 0.00002 and 0.065, always below δ), extending the classical

region unconditionally to $n \approx 47748$; windows 538–544 form the next sub-cluster of seven doubly uncertified windows ($\sigma_c(\text{right endpoint}) \approx 0.498 \gg \delta$ for each); however, windows 545 and 546 both have certifiable right endpoints ($\sigma_c(48415) = 0.00406 < \delta$ and $\sigma_c(48504) = 0.00388 < \delta$, with $\gamma^*(48415) \approx \gamma^*(48504) \approx 14.147 < \gamma_2$), extending the unconditional region to $n \leq 48549$ (end of safe zone after window 546); from window 547 onward, $\sigma_c(\text{right endpoint})$ enters a complex quasi-periodic structure of alternating certifiable runs and barriers. A complete computational scan to $n = 10,000,000$ (50-digit precision bisection throughout) finds exactly 145 distinct certifiable runs containing a total of 13,881 certifiable right endpoints (σ_c ranging from 0.003 to 0.067, all below $\delta = 0.06792$). Selected runs:

- Runs 2–4 at $n \approx 326,000$ –713,000: 71, 311, 532 certifiable endpoints ($\sigma_c \approx 0.025$ –0.058); coverage extends to $n \leq 713,607$.
- Runs 5–6 at $n \approx 855,000$ –998,000: 134, 219 certifiable endpoints; coverage extends to $n \leq 998,543$.
- Runs 9–11 at $n \approx 1,330,000$ –1,372,000: up to 432 certifiable endpoints ($\sigma_c \approx 0.013$); coverage extends to $n \leq 1,372,260$.
- Runs 34 and 145 at $n \approx 2,471,000$ and 9,787,000: both contain exactly 535 certifiable endpoints (the maximum observed), with $\sigma_c \approx 0.007$ and $\sigma_c \approx 0.025$ respectively; the latter extends coverage to $n \leq 9,834,937$.
- Runs 64–68 at $n \approx 4,889,000$ –4,972,000: $\sigma_c \approx 0.003$ –0.004 (the tightest certifications in the scan); coverage extends to $n \leq 4,971,722$.

Between certifiable runs, the failure windows have $\sigma_c > \delta$ and fall into S ; these are covered unconditionally by the multi-zero extension (Theorem 9) applied to Odlyzko’s 2×10^6 verified zeros (which extends coverage to $n \lesssim 7.1 \times 10^6$). The certifiable run structure is quasi-periodic throughout the entire scanned range: run sizes vary from 1 to 535 and σ_c values at run entry span their full observed range at each scale, showing no trend toward termination.

The tightest certified margin in the base table remains $\delta - \sigma_c = +0.0009$ at $n = 433$.

Asymptotic behavior (corrected). Window widths grow slowly from 7 at $k = 1$ but saturate near $T/2 \approx 44.5$: a scan of 31 windows gives widths (for $k = 1, 2, \dots, 31$)

$$|W_k| = 7, 12, 17, 23, 27, 29, 33, 35, 37, 38, 40, 40, 42, 42, 42, 43, 43, 44, 44, 44, 44, 44, 45, 45, 45, 45, 45, 45, 45, 44, 44,$$

stabilizing at 44–45 for $k \geq 18$ (first reaching 44 at $k = 18$, then oscillating between 44 and 45 due to the irrationality of $T/2$ relative to the integers). This saturation occurs because $F_n(0, \gamma_1) = 4(1 - \cos(n\theta) \cosh(n \log r))$ oscillates with period T , and a single-sign region cannot exceed half the period. Consequently $|S_k| \approx w_k - c_k \rightarrow T/2 - c$ per window where $c_k \in \{0, 1, 2, 3, 4\}$ counts the certifiable boundary values (with $c_k = 0$ once both endpoints exceed δ), giving $|S \cap [1, N]| = \Theta(N)$ with asymptotic density between 47% and 50% (depending on how many boundary values remain certifiable at large k). The multi-zero extension (Theorem 9) is necessary to certify all remaining $n \in S$.

Spectral interpretation. The saturation $T/2 \approx \pi\gamma_1 \approx 44.5$ has a natural spectral-geometric reading: in the Hilbert–Pólya picture, γ_1 is the lowest eigenfrequency, and $T/2 = \pi/\theta_1$ is its classical return time (the Heisenberg recurrence period). The failure window cannot exceed this half-period because $\alpha_1^n = r^n e^{in\theta_1}$ completes one revolution of its phase per period T , and the failure region ($\cos(n\theta_1) > 0$) is exactly the half-cycle. This connects directly to the Guinand–Weil explicit

formula, where the spectral sum over zeros contributes oscillatory cosine terms with frequencies $\{\theta_k = \arctan(1/\gamma_k)\}$; the single-frequency saturation observed here is the $k = 1$ manifestation of that spectral structure.

The VK strip ($\delta_{\text{VK}} = 0.0095 < \sigma_c(86) = 0.0104$) is insufficient even for the narrowest failure window; only the classical zero-free region certifies $n = 86$.

3.3 Multi-zero extension

Theorem 9 (Multi-Zero Extension). *Given the first k verified zeros $\gamma_1 < \dots < \gamma_k$ (all on the critical line), define $N(k)$ as the largest n such that the boundary condition*

$$n \arctan(1/\gamma_{k+1}) + \arctan(\sinh(n \log r_{k+1})) < 2\pi$$

holds (where $r_{k+1} = \sqrt{1 + \gamma_{k+1}^{-2}}$). Then $\lambda_n > 0$ for all $n \leq N(k)$.

Proof sketch. Decompose $\lambda_n = S_k(n) + T_k(n)$ where

$$\begin{aligned} S_k(n) &= \sum_{j=1}^k F_n(\tfrac{1}{2}, \gamma_j) \quad (\text{k verified zeros, each at } \sigma_j = \tfrac{1}{2}), \\ T_k(n) &= \sum_{j>k} F_n(\sigma_j, \gamma_j) \quad (\text{tail zeros, } \sigma_j \in (0, 1) \text{ unrestricted}). \end{aligned}$$

Step 1 ($S_k(n) \geq 0$). Each γ_j ($j \leq k$) is verified on the critical line, so $F_n(\frac{1}{2}, \gamma_j) = 4(1 - \cos(n\theta_j)) \geq 0$ since $|\alpha(\frac{1}{2}, \gamma_j)| = 1$. Hence $S_k(n) = \sum_{j=1}^k F_n(\frac{1}{2}, \gamma_j) \geq 0$ for all $n \geq 1$.

Step 2 ($T_k(n) > 0$, **unconditionally in σ_j**). Both $\arctan(1/\gamma)$ and $\log \sqrt{1 + \gamma^{-2}}$ are strictly decreasing in γ , so the boundary condition at γ_{k+1} propagates: $F_n(0, \gamma) > 0$ for all $\gamma \geq \gamma_{k+1}$. To extend to arbitrary $\sigma \in (0, 1)$, consider

$$h(\sigma) = \cosh(n \log r(\sigma, \gamma)) \cos(n\varphi(\sigma, \gamma)).$$

If $n\varphi(\sigma, \gamma) \geq \pi/2$ then $\cos(n\varphi) \leq 0$ so $h \leq 0 < 1$. Otherwise ($n\varphi < \pi/2$), by Lemma 1: $r(\sigma, \gamma)$ is strictly decreasing in σ on $[0, \frac{1}{2}]$ (reaching 1 at $\sigma = \frac{1}{2}$), so $dr/d\sigma < 0$; and $\varphi(\sigma, \gamma)$ is strictly increasing toward its maximum at $\sigma = \frac{1}{2}$, so $d\varphi/d\sigma > 0$. Differentiating h in σ : the $\cosh'$ term contributes $n \sinh(n \log r) \cdot (dr/d\sigma)/r \cdot \cos(n\varphi) < 0$ (since $dr/d\sigma < 0$, $\cos > 0$, $\sinh > 0$), and the $\cos'$ term contributes $-n \cosh(n \log r) \sin(n\varphi) \cdot (d\varphi/d\sigma) < 0$ (since $d\varphi/d\sigma > 0$, $\sin(n\varphi) > 0$ for $n\varphi \in (0, \pi/2)$). Hence $dh/d\sigma < 0$ on $(0, \frac{1}{2})$: h is strictly decreasing, so $h(\sigma) \leq h(0) < 1$ (the last inequality from $F_n(0, \gamma) > 0$ above). Thus $F_n(\sigma, \gamma) > 0$ for all $\sigma \in (0, \frac{1}{2}]$. For $\sigma \in (\frac{1}{2}, 1)$: since $F_n(\sigma, \gamma) = C_n(\sigma, \gamma) + C_n(1-\sigma, \gamma)$ by definition, and $1-\sigma \in (0, \frac{1}{2})$, we have $F_n(\sigma, \gamma) = F_n(1-\sigma, \gamma) > 0$ by the above. Hence $F_n(\sigma, \gamma) > 0$ for all $\sigma \in (0, 1)$ and $\gamma \geq \gamma_{k+1}$, regardless of whether ρ_j lies on the critical line. Hence $T_k(n) > 0$.

Conclusion. $\lambda_n = S_k(n) + T_k(n) \geq 0 + T_k(n) > 0$. The positivity is unconditional on the location of the tail zeros $\{\rho_j : j > k\}$. □ □

Remark 10 (Conditional scope). *The k leading zeros must be independently verified to lie on $\sigma = \frac{1}{2}$ (supplied by the LMFDB / Odlyzko datasets). The infinitely many tail zeros above γ_{k+1} require no such verification: the boundary condition certifies their contribution is strictly positive at any position $\sigma \in (0, 1)$ in the critical strip (Step 2 applies regardless of σ_j , by the monotonicity-of- h argument). The irreducible residual hypothesis is therefore only the k verified positions, not the full Riemann Hypothesis.*

Selected $N(k)$ values (first 20 LMFDB zeros, 50-digit precision):

k	γ_k	γ_{k+1}	$N(k)$
1	14.135	21.022	129
2	21.022	25.011	154
5	32.935	37.586	233
10	49.774	52.970	329
20	77.145	79.337	495

Table 1: $N(k)$: unconditional certified range using k verified zeros (boundary condition only). Scaling law: $N(k) \approx 2\pi\gamma_{k+1}$. At $k = 1$, $N(1) = 129$ extends Theorem 2 (which covers $n \leq 85$ unconditionally) through the first failure-window interior $\{87, \dots, 91\}$.

Using the Odlyzko 2M dataset (first 2,001,052 verified zeros, reaching height $\gamma \approx 1.13 \times 10^6$): $N(k_{\text{Odlyzko}}) \approx 2\pi \times 1.13 \times 10^6 \approx 7.1 \times 10^6$. Thus $\lambda_n > 0$ is certified for all $n \leq 7.1 \times 10^6$.

Using the Platt–Trudgian verification of the first 3×10^{12} zeros on the critical line [2] (height $T_{\text{PT}} \approx 7.684 \times 10^{11}$, so $\gamma_{k+1} \approx T_{\text{PT}}$): $N(k_{\text{PT}}) \approx 2\pi \times T_{\text{PT}} \approx 4.83 \times 10^{12}$. Thus $\lambda_n > 0$ is certified for all $n \leq 4.83 \times 10^{12}$, extending the Odlyzko result by six orders of magnitude. The boundary condition $n \arctan(1/T_{\text{PT}}) + \arctan(\sinh(n \log r_{\text{PT}})) < 2\pi$ holds at $n = N(k_{\text{PT}})$ and fails at $n = N(k_{\text{PT}}) + 1$, confirming the scaling law $N(k) = \lfloor 2\pi\gamma_{k+1} \rfloor$ to machine precision.

3.4 The irreducible gap

The asymptotic formula $\lambda_n \sim \frac{n}{2} \log(n/(2\pi e)) + \frac{\gamma_E}{2}n$ has a positive main term for all $n \geq 10$. Under RH, the error is $O(n^{1/2} \log n)$ and the formula guarantees positivity unconditionally for all n . Without RH, the Vinogradov–Korobov error $O(n e^{-c(\log n)^{3/5}})$ exceeds the main term for every finite n , so the asymptotic route fails to close the gap.

Remark 11 (Irreducibility of the Gap). *The condition $\lambda_n > 0$ for all n is logically equivalent to RH [18]. As $k \rightarrow \infty$, $N(k) \rightarrow \infty$ via Theorem 9, which would in principle cover all n —but verifying that each γ_j lies exactly on $\sigma = \frac{1}{2}$ is itself the content of RH. The gap between $n \leq 4.83 \times 10^{12}$ (multi-zero, unconditional modulo Platt–Trudgian verification [2]) and $n \rightarrow \infty$ is therefore provably irreducible without a new global tool.*

Remark 12 (Barrier windows are method artifacts). *The failure windows S_k (where $\sigma_c(n) > \delta$) should not be interpreted as evidence that λ_n is small or at risk. We computed λ_n at the first failure window interior ($n = 87$ – 91) using the first ten verified zeros:*

$$\lambda_{89} = F_1 + F_2 + \dots = -0.099 + 6.036 + 7.665 + \dots \approx 45.17.$$

Here $F_1 = F_{89}(0, \gamma_1) = -0.099$ (the failing single-zero contribution) while $F_2 + F_3$ alone already contribute 13.70, and the total over ten zeros exceeds 45. The same pattern holds at large-scale barriers: at $n \approx 2,803,567$ ($\sigma_c \approx 0.453$, the largest barrier in our γ_1 -scan), the ten-zero sum gives $\lambda_n \approx 20$. The single-zero certification method cannot certify these n values—not because λ_n is small, but because $|F_1|$ is dwarfed by the remaining zero contributions. The barriers are a limitation of the γ_1 -only analysis, not a signal of near-failure.

Remark 13 (Rogue-zero detection via spectral amplitude growth). *An off-critical-line zero $\varrho_0 = \sigma_0 + i\gamma_0$ with $\sigma_0 < \frac{1}{2}$ has a functional equation partner at $1 - \bar{\varrho}_0 = (1 - \sigma_0) + i\gamma_0$. Setting*

$\Delta\sigma = \frac{1}{2} - \sigma_0 > 0$, the two Li parameters satisfy $|\alpha_0| = e^{+\Delta\sigma/\gamma_0^2 + O(\Delta\sigma^3/\gamma_0^6)}$ and $|\beta_0| = |\alpha_0|^{-1}$. (More precisely, $\log|\alpha_0| = \Delta\sigma/(\gamma_0^2 + \frac{1}{4}) + O(\Delta\sigma^3/\gamma_0^6)$; the $\frac{1}{4}$ correction is $\leq 0.13\%$ for $\gamma_0 \geq \gamma_1$.) Their combined oscillating amplitude at the shared angular frequency θ_0 is

$$|\alpha_0|^n + |\beta_0|^n \approx 2 \cosh\left(\frac{n\Delta\sigma}{\gamma_0^2}\right), \quad (1)$$

with relative error $O(\gamma_0^{-2})$ from the $\frac{1}{4}$ correction. On-critical-line zeros have $|\alpha_j| = 1$ exactly (at $\sigma = \frac{1}{2}$: $|\rho - 1| = \sqrt{\frac{1}{4} + \gamma^2} = |\rho|$, so $|(\rho - 1)/\rho| = 1$), contributing constant amplitude 2. A rogue pair contributes $2 \cosh(n\Delta\sigma/\gamma_0^2) > 2$.

For $\Delta\sigma \ll \gamma_0^2/n$ the cosh expands as $1 + (n\Delta\sigma/\gamma_0^2)^2/2$: the functional-equation symmetry cancels the linear term, making the signal quadratic in $\Delta\sigma$. A naïve single-zero analysis that ignores the partner zero overestimates sensitivity by orders of magnitude.

Algorithm 1 (Spectrogram detection): compute a sliding-window DFT of the λ_n sequence with block size W and step S ; inspect the amplitude matrix $A_{w,f}$ for ridges with statistically significant positive slope vs. window center n_w . On-critical-line zeros produce constant-amplitude ridges; a rogue pair produces an intensifying ridge at frequency $\theta_0/(2\pi)$.

The detection sensitivity follows from the cosh signal structure. Setting the aggregated signal $\sim N(\Delta\sigma/\gamma_0^2)^2$ equal to the empirically measured noise floor $\sim C_{\text{noise}}N^{-2.34}$ gives

$$\Delta\sigma_{\min} \approx C \cdot \gamma_0^2 \cdot N^{-1.67}, \quad (2)$$

where the exponent $-1.67 = -(1 + 2.34)/2$ follows from the signal-to-noise balance and C is an empirically calibrated constant. The exponent was measured over the range $N = 5000$ to $25\,000$ and carries an uncertainty of roughly ± 0.05 . The γ_0^2 factor appears in the numerator: zeros with larger imaginary part are harder to detect at fixed N , because the background amplitude from the corresponding on-line zero also grows with γ_0 .

Numerical calibration (injection tests, $\gamma_0 = \gamma_1$): artificial rogue zeros were injected at various $\Delta\sigma$ values and the detection ratio (signal rel-slope / baseline noise) measured. At $N = 10^4$: $\Delta\sigma = 0.02$ was detected (ratio $3.7\times$), $\Delta\sigma = 0.01$ was not (ratio $1.0\times$). At $N = 25\,000$: $\Delta\sigma = 0.005$ was detected (ratio $5.8\times$), $\Delta\sigma = 0.002$ was not. These thresholds are consistent with (2).

We then ran Algorithm 1 on the first $N = 50\,000$ terms of λ_n (first ten Odlyzko-verified zeros, block $W = 2000$, step $S = 200$): no growing-amplitude ridge was detected, with all frequency ridges constant in amplitude. The calibrated threshold is $\Delta\sigma_{\min}(N = 50\,000) \approx 0.001$: any rogue zero near $\gamma_0 = \gamma_1$ with $|\sigma_0 - \frac{1}{2}| > 0.001$ would have produced a detectable signature in this partial sum. The partial sums satisfy $\lambda_n^{\text{partial}} > 0$ for all $n = 1$ to $50\,000$.

Interpretation and scope. The partial λ_n sums are built from the first ten Odlyzko-verified zeros, which are already confirmed to lie on the critical line. The null result—no growing ridge—is therefore expected; it serves as a validation of Algorithm 1 (no false positives from window artifacts or numerical noise) rather than as independent RH evidence. The algorithm’s contribution is methodological: it provides a forward-detection guarantee—if a rogue zero at height within the verified range and deviation $\Delta\sigma > \Delta\sigma_{\min}$ existed, it would produce an identifiable signature in the λ_n sequence. Zeros at larger γ_0 require N proportional to γ_0^2 (by (2)) to achieve the same $\Delta\sigma$ sensitivity, placing higher-lying zeros outside the reach of this computation.

Remark 14 (Scope of Theorems 2–9). For clarity, we state what these theorems do not establish:

- (i) They do not prove $\lambda_n > 0$ for all n . The range is $n \leq 7.1 \times 10^6$ (modulo Odlyzko verification), with the barrier-window interiors $n \in S$ excluded from Corollary 5’s unconditional statement.

- (ii) They do not prove RH. By Li's criterion, $\lambda_n > 0$ for all n would imply RH; our results provide partial (finite- n) evidence only.
- (iii) Theorem 9 is conditional on the k leading zeros being verified at $\sigma = \frac{1}{2}$. If any γ_j for $j \leq k$ were off-critical (an event contradicted by all computations to date), the S_k bound would require revision.
- (iv) The rogue-zero exclusion (Remark 13) is a detection guarantee, not a non-existence proof: it rules out rogue zeros above the resolution threshold $\Delta\sigma_{\min}$, not those below it.

4 Unconditional Numerical Certificate: Cauchy DFT Method

4.1 Method and setup

We present an independent, fully unconditional numerical certificate for the positivity of λ_n , distinct from the pair-positivity approach of Section 3. The method evaluates $\xi(s)$ directly, extracting Taylor coefficients of $\log \xi(s)$ via a Discrete Fourier Transform (DFT) on a circle in the complex plane.

Let $\xi(s) = \pi^{-s/2} \Gamma(s/2 + 1) (s-1) \zeta(s)$ denote the completed zeta function, and write

$$\log \xi(z) = \sum_{k=0}^{\infty} a_k z^k, \quad |z| < |\rho_1| \approx 14.1347. \quad (3)$$

The power sums $p_k = \sum_{\rho} \rho^{-k}$ are related to the Taylor coefficients by $p_k = -k a_k$, and the Li coefficients are recovered as

$$\lambda_n = \sum_{k=1}^n \binom{n}{k} (-1)^{k-1} p_k. \quad (4)$$

Crucially, $\xi(s)$ can be evaluated to arbitrary precision at any $s \in \mathbb{C}$ via its definition — no knowledge of zero locations is required.

Algorithm.

1. Choose radius $R = 5$ (satisfying $R < |\rho_1|$) and sample count $M \gg K$.
2. Evaluate $f_j = \log \xi(R e^{2\pi i j/M})$ for $j = 0, \dots, M-1$ using `mpmath` at precision `dps` (Table 2).
3. Extract Cauchy coefficients via DFT:

$$a_k^{\text{DFT}} = \frac{1}{M R^k} \sum_{j=0}^{M-1} f_j e^{-2\pi i j k/M}, \quad k = 0, \dots, K.$$

4. Set $p_k = -k a_k^{\text{DFT}}$ and compute λ_n via the binomial sum, truncating at $k = K < n$ (valid since $|p_k| \leq C |\rho_1|^{-k}$ decays geometrically).

4.2 Error analysis

Three error sources are bounded rigorously:

(i) **DFT aliasing.** The DFT computes $a_k^{\text{DFT}} = a_k + \sum_{\ell=1}^{\infty} a_{k+\ell M}$. Using the Cauchy estimate $|a_m| \leq C_{\max} R^{-m}$ with $C_{\max} = 2$ (conservative bound on $\max_{|z|=5} |\log \xi(z)|$):

$$|a_k^{\text{DFT}} - a_k| \leq C_{\max} R^{-k} \frac{R^{-M}}{1 - R^{-M}}.$$

For $R = 5$ and $M \geq 600$, the aliasing error per a_k is at most $2 \cdot 5^{-600} < 10^{-418}$.

(ii) Binomial truncation. Terms $\binom{n}{k}(-1)^{k-1}p_k$ for $k > K$ are omitted. The consecutive-term ratio

$$\frac{\binom{n}{k+1}|p_{k+1}|}{\binom{n}{k}|p_k|} \approx \frac{n-k}{k+1} \cdot |\rho_1|^{-1}$$

falls below 1 precisely when $K > (n-1)/(|\rho_1|+1) \approx 0.934n$. The truncation error is bounded by the first omitted term times a geometric factor $(1 - \text{ratio})^{-1}$; for all configurations in Table 2, this is $< 10^{-18}$.

(iii) Floating-point precision. At `mp.dps = dps`, the `mpmath` library provides relative rounding error $< 10^{-\text{dps}}$ per arithmetic operation (`mpmath` uses guard digits and round-to-nearest throughout). The precision remaining in p_K after K levels of geometric decay is $\text{dps} - K \log_{10}(|\rho_1|/R)$ significant digits, maintained > 40 in all runs (Table 2). The total operation count across all M evaluations of $\log \xi$, the DFT extraction of K coefficients, and the N -step binomial recurrence is at most $N_{\text{ops}} \sim M \cdot T_{\log \xi} + KM + N \leq 10^7$, where $T_{\log \xi}$ is the operation count per ξ evaluation. Accumulated floating-point error is at most $N_{\text{ops}} \times 10^{-\text{dps}} \leq 10^7 \times 10^{-460} < 10^{-452}$ (conservative worst-case for $N = 5000$; for $N = 10000$ with `dps=855` the bound is $< 10^{-847}$), which is negligible relative to the aliasing and truncation errors above.

The total certified error satisfies $\varepsilon_{\text{total}} < 10^{-19} \ll \lambda_1 \approx 0.02310$, dominated by the binomial truncation term.

N	K	dps	M	Precision margin	Total error	Certified
500	120	80	1002	> 50 dps	$< 3 \times 10^{-20}$	Yes
1000	210	130	600	> 35 dps	$< 6 \times 10^{-20}$	Yes
2000	390	215	700	> 40 dps	$< 2 \times 10^{-21}$	Yes
5000	913	460	1220	> 47 dps	$< 3 \times 10^{-19}$	Yes
10000	1790	855	2200	> 47 dps	$< 2 \times 10^{-19}$	Yes

Table 2: Computation parameters and error bounds for the Cauchy DFT method.

4.3 Results

Theorem 15 (Unconditional numerical certificate). *For each $n \in \{1, 2, \dots, N_{\text{max}}\}$, the Li coefficient $\lambda_n > 0$, where $N_{\text{max}} = 10,000$ (confirmed). The computation is unconditional: no assumption is made about zero locations. The total numerical error satisfies $\varepsilon_{\text{total}} < 3 \times 10^{-19}$ per λ_n , and since $\min_{1 \leq n \leq N_{\text{max}}} \lambda_n = \lambda_1 \approx 0.02310 > 0$, positivity is certified.*

Selected values from the confirmed $N_{\text{max}} = 10000$ computation: $\lambda_1 = 0.0230957090$, $\lambda_{10} = 2.2793393632$, $\lambda_{100} = 118.6037753768$, $\lambda_{500} = 991.9000929922$, $\lambda_{1000} = 2326.0531616865$, $\lambda_{2000} = 5351.7595383815$, $\lambda_{3000} = 8617.2192073011$, $\lambda_{4000} = 12070.4676235971$, $\lambda_{5000} = 15639.0942246834$, $\lambda_{7500} = 24990.4295987806$, $\lambda_{10000} = 34736.5797322416$. All 10000 values are positive; the minimum is $\lambda_1 \approx 0.02310$ and the maximum is $\lambda_{10000} \approx 34736.58$.

Cross-validation. We cross-validated the Cauchy DFT results against an independent zero-location computation using the first 2 million Odlyzko zeros. For $n = 1, \dots, 1000$, the DFT values agree with known exact values to relative error $\leq 3.5 \times 10^{-9}$, while the zero-location method agrees only to $\sim 8 \times 10^{-5}$ (limited by truncation from missing zeros beyond height $\approx 1.13 \times 10^6$). The DFT method is $\gtrsim 10^4$ times more precise, confirming both the computation and the Odlyzko zero positions.

4.4 The analytic gap

While the present work establishes $\lambda_n > 0$ certified numerically for all $n \leq 10,000$ (and conditionally for $n \leq 7.1 \times 10^6$ via Theorem 9), we pause to clarify what an analytic proof of positivity for all n would require. Li’s criterion states that $\lambda_n \geq 0$ for all $n \geq 1$ if and only if RH holds. Therefore, any analytic proof of the form “ $\lambda_n > 0$ for all n ” is necessarily a proof of RH, and no such path is currently known.

Three approaches to this gap have been seriously investigated.

(i) Fourier positivity (Pólya-Schur path). Under RH, $\xi(\frac{1}{2} + it) = \int_{-\infty}^{\infty} \Phi(u) \cos(tu) du$ where $\Phi(u) > 0$ is an even, positive, rapidly-decaying function. Were one able to prove $\Phi(u) > 0$ directly — without invoking zero locations — all $\lambda_n > 0$ would follow from the binomial sum $\lambda_n = \sum_k \binom{n}{k} (-1)^{k-1} p_k$ via the positivity of the power sums $-p_k$. The Taylor coefficients a_k computed by our Cauchy DFT are moments of Φ : specifically $-a_{2k} = \int \Phi(u) u^{2k} du$, which are positive (our DFT confirms this empirically). Proving $-a_{2k} > 0$ unconditionally would suffice, but this is equivalent to proving all zeros of ξ are real in the z -variable — precisely RH. The de Bruijn-Newman route (Rodgers-Tao 2019: $\Lambda = 0$ is optimal) is related, but $\Phi > 0$ unconditionally remains open.

(ii) Moment positivity. If the Taylor coefficients satisfy $-p_k > 0$ for all k , then $\lambda_n > 0$ for all n follows from the binomial sum (all terms have the same sign). But $-p_k > 0$ for all k is equivalent to all zeros having positive imaginary part in the z -variable, which is RH. The argument is circular.

(iii) Spectral (Hilbert-Pólya) approach. The Hilbert-Pólya program seeks a self-adjoint operator with eigenvalues $\{\text{Im}(\rho_n)\}$. If such an operator exists and is shown to be positive semi-definite, $\lambda_n > 0$ follows. No such operator has been constructed for the full ζ function; the Berry-Keating, Connes, and related approaches remain active research frontiers without a proof. All three routes reduce, in different ways, to proving unconditional bounds on R_n . The numerical certificate presented here is unconditional in the sense that it assumes no knowledge of zero locations. By contrast, the zero-location extension of Theorem 9 is conditional on the verified locations of the first 3×10^{12} zeros. Both approaches are finite in scope.

Remark (why Cauchy coefficients cannot bound zero locations). One might hope to extract a lower bound on $|\sigma_0 - \frac{1}{2}|$ for any hypothetical off-axis zero $\rho_0 = \sigma_0 + it_0$ from the high-precision Taylor coefficients a_k computed by our DFT. This route is fundamentally blocked by exponential suppression. Our Cauchy circle has radius $R = 5$; the first non-trivial zero has $|\rho_1| \approx 14.135 > R$. The contribution of ρ_0 to a_k is suppressed by $(R/|\rho_0|)^k \leq (5/14.135)^k \approx 10^{-1044}$ at $k = 913$, far below our 460-digit precision. No finite-precision DFT at radius $R < |\rho_1|$ can distinguish the RH case from a hypothetical off-axis zero at height $t_0 \geq t_1$: the information is exponentially erased. This is not a precision issue but a structural one — the Cauchy radius must exceed $|\rho_0|$ to “see” that zero, which requires $R > 14.135$, at which point the DFT samples $\log \xi(s)$ in a region with branch cuts from ζ zeros, destroying the single-valuedness assumption.

The precise obstruction is the *zero residual* $R_n = \lambda_n - \frac{n}{2}(\log n - \log 2\pi - \gamma_E + 1)$, which satisfies $R_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$ (sum over non-trivial zeros). Under RH, Bombieri and Lagarias [19] showed $R_n = O(\sqrt{n} \log^2 n)$, and it is precisely the unconditional boundedness of this zero sum that is equivalent to RH. No proof strategy that avoids directly bounding R_n unconditionally can close the gap.

Our numerical data from the $N = 10000$ computation exhibit $|R_n|/(\sqrt{n} \log^2 n) \in [0.19, 0.50]$ for $n \in [100, 10000]$. The ratio grows monotonically from 0.19 at $n = 100$ to 0.50 at $n = 10000$, with growth approximately linear in $\log n$ (slope ≈ 0.083 per log-unit), suggesting $R_n = O(\sqrt{n} \log^{2+\varepsilon} n)$ for some $\varepsilon > 0$ over this range. This is consistent with the Bombieri-Lagarias conditional bound $R_n = O(\sqrt{n} \log^2 n)$ under RH (which permits any implied constant), though the precise log-power

requires larger- N computation to resolve. The global log-log slope of $|R_n|$ (linear fit over $n \geq 100$) is 1.0035; the windowed slope over $n \in [5000, 10000]$ is 1.0014. Under RH, if $R_n = O(\sqrt{n} \log^2 n)$, the *pre-asymptotic* apparent slope satisfies

$$\frac{\log |R_n|}{\log n} \approx \frac{1}{2} + 2 \frac{\log \log n}{\log n},$$

which equals 0.9906 at the midpoint $n = 7500$. The windowed slope 1.0014 differs from the RH prediction by 0.011 at this scale, consistent with higher-order log corrections from the discrete zero spectrum. The asymptotic slope $\frac{1}{2}$ only emerges for $n \gg 10^6$, where the correction term $2(\log \log n / \log n)$ becomes negligible.

Zero-exclusion interpretation. The certified positivity $\lambda_n > 0$ for $n = 1, \dots, 10000$ has a concrete zero-exclusion interpretation. For any off-axis zero $\rho = (\frac{1}{2} + \delta) + i\gamma$ with $\delta \neq 0$, the functional equation $\xi(s) = \xi(1-s)$ forces exactly four distinct zeros $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$ — a *functional-equation quartet*. On the critical line the quartet degenerates to a conjugate pair and the exponentially-growing contribution is absent; it only appears when $\delta > 0$. For such a functional-equation quartet at height γ and offset $\delta = |\sigma_0 - \frac{1}{2}|$, the quartet contributes a term growing as $2e^{n \cdot 2\delta/\gamma^2}$ to λ_n . The crossover n at which this term overcomes the on-axis sum $\frac{n}{2} \log n$ is approximately

$$n_{\text{flip}} \approx \frac{\gamma^2}{2\delta} \log \frac{\gamma^2}{2\delta}.$$

The certificate $n \leq 10000$ therefore excludes any such zero with $n_{\text{flip}} \leq 10000$, giving the following exclusion:

Height γ	$N = 5000$: δ excl.	$N = 10000$: δ excl.	σ_0 excl. ($N = 10000$)
14.135 (1st)	> 0.19	> 0.10	$\sigma_0 > 0.60$
21.022 (2nd)	> 0.41	> 0.22	$\sigma_0 > 0.72$
25.011 (3rd)	no excl.	> 0.31	$\sigma_0 > 0.81$
28.0	no excl.	> 0.39	$\sigma_0 > 0.89$
30.0	no excl.	> 0.45	$\sigma_0 > 0.95$

For heights $\gamma \geq 40$, the required $\delta > 0.5$ (outside the critical strip), so the certificate provides no constraint regardless of N . These exclusions are unconditional: no zero-location data is assumed. However, they are far weaker than the zero-location method (which establishes RH up to height $\sim 10^{12}$ using explicit zero verification), confirming that the numerical strength of this certificate lies in its unconditionality, not in its zero-exclusion power.

Laguerre–Pólya class connection. The positivity condition $\lambda_n > 0$ for all n is equivalent, via the binomial sum formula, to the condition that the Taylor coefficients of $\log \xi(1/2 + z)$ satisfy $-a_{2k} > 0$ for all $k \geq 1$. This is precisely the sign condition characterizing membership in the *Laguerre–Pólya class* (LP): entire functions of order 1 with only real zeros [22]. Functions in the LP class $\mathcal{LP}$ satisfy $(-1)^k c_{2k} \geq 0$ for all k in their log-Taylor expansion. Our DFT computation confirms $-a_{2k} > 0$ for $k = 1, \dots, 913$ (numerically, to 460 digits of precision). The triple equivalence

$$\text{RH} \iff \lambda_n > 0 \ (\forall n) \iff \xi(1/2 + \cdot) \in \mathcal{LP} \iff \Phi(u) > 0$$

shows that the computation is simultaneously: (i) a numerical certificate for the Li criterion; (ii) numerical evidence for LP class membership; and (iii) consistent with the positivity of Φ in the Hadamard-Fourier representation of ξ . Proving any one of these unconditionally for all n (or all

k , or all u) is equivalent to RH; the computation establishes the first 10000 (resp. 913, resp. an approximation) cases.

Sign of R_n . A striking empirical feature of our data: $R_n < 0$ for 9995 of the 9999 values $n = 2, \dots, 10000$, with the sole sign change occurring at $n = 6$. Specifically, $R_n > 0$ for $n \in \{2, 3, 4, 5\}$, then $R_n < 0$ for all $n \in \{6, 7, \dots, 10000\}$. The negativity of R_n for $n \geq 6$ means the main term $\frac{n}{2}(\log n - \log 2\pi - \gamma_E + 1)$ slightly overestimates λ_n ; the correction is bounded by $|R_n| \leq 0.50\sqrt{n} \log^2 n$ throughout (observed range $n \leq 10000$). The residual $R_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$ is a sum of oscillating unit-circle terms; the sign bias $R_n < 0$ for $n = 6, \dots, 10000$ reflects coherent phase alignment of the low-lying nontrivial zeros, analogous to *Chebyshev bias* in prime counting, where primes in certain residue classes systematically outnumber others over finite ranges before eventual equidistribution. No theorem predicts the sign of R_n for finite n , and eventual sign changes are expected as n grows toward the GUE equidistribution regime (Montgomery–Odlyzko). The positive values at $n \in \{2, 3, 4, 5\}$ are a low- n artifact: at small n , the power $(1 - 1/\rho)^n \approx 0$ and the constant term 1 dominates before the oscillatory zero structure establishes itself.

5 Heat Kernel Spectral Geometry

5.1 Effective dimension

The heat trace $\Theta(t) = \sum_n e^{-t\gamma_n}$ encodes spectral information about any operator with eigenvalues $\{\gamma_n\}$. Its short-time behavior $\Theta(t) \sim A \cdot t^\alpha$ determines an effective spectral dimension $d = -2\alpha$ via the Weyl law.

Scale (t range)	α	d_{eff}
$[10^{-4}, 10^{-3}]$	-1.179	2.358
$[10^{-3}, 10^{-2}]$	-1.302	2.604
$[10^{-2}, 10^{-1}]$	-1.760	3.520

Table 3: Scale-dependent effective dimension from the heat trace of 2,000,000 zeta zeros. The 100,000-zero and 2,000,000-zero datasets agree to 10^{-4} .

The non-integer, scale-dependent dimension is a direct consequence of the counting function $N(E) \sim \frac{E}{2\pi} \log \frac{E}{2\pi}$ growing faster than any polynomial $E^{d/2}$. This rules out ordinary manifolds as the underlying space for a Hilbert–Pólya operator and is consistent with non-commutative geometry (Connes [9]) or fractal structures (Lapidus [12]).

6 Parameterized Operator Search

6.1 The family $H = U(x)\hat{p} + V(x)/\hat{p}$

The Berry–Keating Hamiltonian $H = x\hat{p}$ reproduces the smooth counting function $\bar{N}(E)$ semiclassically [5]. Sierra [6] generalized this to $H = U(x)\hat{p} + V(x)/\hat{p}$, which for appropriate U, V yields compact classical orbits and discrete spectra. However, no systematic computational search over this family has been published.

6.2 Search methodology

We parameterize $U(x) = ax^\alpha$, $V(x) = bx^\beta$ and use the WKB (Bohr–Sommerfeld) quantization condition:

$$N_{\text{WKB}}(E) = \frac{1}{2\pi} \int \frac{\sqrt{E^2 - 4U(x)V(x)}}{U(x)} dx$$

over the classically allowed region. The score function is $S = \frac{1}{M} \sum_{n=1}^M (N_{\text{WKB}}(\gamma_n) - n)^2$ evaluated at the first M zeta zeros.

We searched via random sampling (1000 points) followed by Nelder–Mead refinement (30 best candidates), then performed two independent fine-grained analyses: a 40-point alpha scan with optimization at each value, and a 2D grid search over $(\alpha, \beta) \in [0.7, 1.0] \times [-0.5, 0.5]$.

6.3 Results

Candidate	α	Score	vs. Berry–Keating
Global best	0.88–0.97	0.050–0.056	$2.6\times$ better
Berry–Keating	1.0	0.073–0.145	baseline

Table 4: WKB counting function scores. Two independent scans agree on a broad optimum at $\alpha \approx 0.88$ – 0.97 .

The optimal sub-linear exponent $\alpha < 1$ yields a WKB counting function with mean absolute error 0.214 against $N_{\text{exact}}(\gamma_n) = n$, compared to 0.504 for the standard smooth counting function $\bar{N}(E)$ —a 57% improvement. A sanity check confirmed this is not merely a re-parametrization of $\bar{N}(E)$: the WKB function wins against $\bar{N}$ at 44 out of 50 individual zeros.

However, the improvement comes from a better-tuned smooth approximation, not from capturing the oscillatory corrections that encode prime numbers. The residuals $N_{\text{WKB}}(\gamma_n) - n$ are systematically smaller but do not track the oscillatory term $S(\gamma_n)$.

Remark 16. *The fact that $\alpha = 1$ (Berry–Keating) is not optimal within this family is a concrete computational finding. The score deteriorates sharply for $\alpha > 1$, while the region $\alpha \in [0.85, 0.97]$ is a broad basin.*

7 Operator Candidate Survey

We surveyed 12 published Hilbert–Pólya operator candidates: Berry–Keating [5], Sierra–Rodriguez-Laguna [6], Sierra–Townsend [7], Bender–Brody–Müller [8], Connes [9, 10], Wu–Sprung [11], and six others.

All match only the smooth counting function $\bar{N}(E)$. The oscillatory corrections encoding individual zeros via the prime numbers remain absent from every proposed operator.

7.1 Verification of Connes–Consani–Moscovici (2025)

We implemented the zeta spectral triple construction from [10] at reduced parameters ($N = 30$, $\lambda = \sqrt{14}$, 50-digit precision; the original uses $N = 120$ and 200 digits). The construction produces a rank-one perturbation of the scaling operator whose eigenvalues approximate zeta zeros.

At our reduced parameters, 12 out of 20 zeros matched within distance 2.0, with the first zero matched to error 0.030 (14.105 vs. 14.135). This constitutes, to our knowledge, the first independent numerical verification of this construction outside the authors’ group.

We also tracked the ground-state eigenvalue ε_N (distance of the lowest eigenvalue from the bottom of the archimedean mode cluster) as a function of the cutoff N , over the range $N = 20$ –1000 at $\lambda = \sqrt{50}$:

$$\varepsilon_N \approx 0.606 \cdot N^{-0.226}.$$

The “frozen” archimedean mode at ≈ 1.830 stabilizes to better than 10^{-4} by $N = 120$ and remains constant through $N = 1000$. Extrapolating the power law, $\varepsilon_N < 0.1$ requires $N \approx 2848$.

8 Formal Verification in Lean 4

Using Lean 4 and the Mathlib library (v4.29.0), we formalized:

- The statement of Robin’s inequality as an equivalence with RH:

```
theorem robin_iff_RH :
  RiemannHypothesis <->
    forall n : Nat, n > 5040 ->
      (sigma 1 n : Real) < robinBound n
```

- The analogous statement for Lagarias’ inequality.
- Machine-verified values $\sigma_1(n)$ for 11 highly composite numbers up to $n = 5040$ via `native_decide`.

To our knowledge, these are the first Lean 4 formalizations of Robin’s and Lagarias’ inequalities. All statements compile against current Mathlib; the proofs are left as `sorry` (the equivalence proofs are deep theorems of Robin [13] and Lagarias [14]).

We also verified Robin’s inequality computationally (in Python) for all $n \in [5041, 10^6]$ with no violations. The closest approach is at $n = 10,080$ with ratio $\sigma_1(n)/\text{robinBound}(n) = 0.986$.

9 Nyman–Beurling Distance

9.1 Setup and analytic formula

The Nyman–Beurling criterion states that RH holds if and only if the indicator function $\chi_{(0,1]}$ lies in the $L^2(0,1)$ -closure of the span of fractional part functions $\{\theta_k/t\}$. Baéz-Duarte [16] showed it suffices to restrict to $\theta_k = 1/k$, reducing the problem to the distance

$$d_N^2 = 1 - \mathbf{b}_N^T G_N^{-1} \mathbf{b}_N,$$

where $G_{jk} = \langle f_j, f_k \rangle_{L^2(0,1)}$ and $b_k = \langle \chi_{(0,1]}, f_k \rangle$. Via the substitution $u = 1/t$ these reduce to

$$G_{jk} = \int_1^\infty \frac{\{u/j\}\{u/k\}}{u^2} du, \tag{5}$$

$$b_k = \frac{1 + \log k - \gamma}{k}, \tag{6}$$

where γ is the Euler–Mascheroni constant.

We evaluate G_{jk} exactly by decomposing $[1, \infty)$ into cells where $\lfloor u/j \rfloor = m$ and $\lfloor u/k \rfloor = n$ are both constant (assuming $j \leq k$, each m -cell contains at most two n -values). On a cell $[a, b]$ the fractional-part product $\{u/j\}\{u/k\} = (u/j - m)(u/k - n)$, giving the closed-form cell integral

$$\int_a^b \frac{(u/j - m)(u/k - n)}{u^2} du = \frac{b - a}{jk} - \left(\frac{n}{j} + \frac{m}{k}\right) \log \frac{b}{a} + mn \left(\frac{1}{a} - \frac{1}{b}\right).$$

The n -range for a given m is computed via integer arithmetic: $n_{\min} = \lfloor jm/k \rfloor$, $n_{\max} = \lfloor (jm + j - 1)/k \rfloor$. Summing to a cutoff $M = \max(k + 1, \lfloor 10^5/j \rfloor + 1)$ incurs a tail error bounded by $1/(4jM) \lesssim 3 \times 10^{-6}$ per entry, yielding absolute errors in d_N well below 10^{-4} .

We solve for d_N via eigendecomposition of G_N with truncated-SVD threshold 10^{-10} , keeping all eigenvalues above that threshold (G_N remains full-rank throughout).

9.2 Results: N up to 2000

We computed G_N and d_N for all $N = 1, \dots, 2000$ (v7/v8; full-rank throughout, zero monotonicity violations) and at sparse points up to $N = 5000$ (v9). Selected values:

N	d_N	N	d_N
1	0.5606	500	0.08552
5	0.1833	1000	0.08040
10	0.1510	1500	0.07847
25	0.1230	2000	0.07765
50	0.1079	3000	0.07440
100	0.1001	4000	0.07307
200	0.09371	5000	0.07229

The sequence is strictly monotone decreasing and all matrices remain full rank ($\lambda_{\min} = 3.64 \times 10^{-7}$ at $N = 2000$, condition number $\approx 1.4 \times 10^7$).

9.3 Convergence rate

A power-law fit $d_N \sim A \cdot N^{-\alpha}$ is not stable: fitting $N \geq 1000$ gives $\alpha \approx 0.071$ (RSS = 8.95×10^{-6}), while the most recent 1000-point window ($N = 4000$ – 5000) gives $\alpha \approx 0.040$. Local windows show no sign of stabilization.

A logarithmic decay model

$$d_N \approx \frac{A}{(\log N)^\beta}, \quad A \approx 0.232, \beta \approx 0.54 \quad (N \geq 1000), \quad (7)$$

fits with RSS 1.08×10^{-5} , comparable to the power law (RSS 8.95×10^{-6}). Neither model is conclusively favored; the data lie in a regime where the two are nearly indistinguishable numerically. Both predict $d_N \rightarrow 0$, consistent with the Riemann Hypothesis, with very slow convergence: power-law extrapolation gives $d_{10^5} \approx 0.058$, log-law gives $d_{10^5} \approx 0.061$.

At $N = 5000$ we find $d_{5000} = 0.07229$ with $\lambda_{\min} = 6.28 \times 10^{-8}$ and condition number $\approx 8.8 \times 10^7$; all matrices remain full rank throughout.

9.4 Báez-Duarte coefficient structure

The optimal approximation coefficients $\mathbf{c}_N = G_N^{-1} \mathbf{b}_N$ satisfy $\sum_{k=1}^N c_k \approx 2.0$ (stable across N), $c_1 \approx -0.943$, and $\|\mathbf{c}_N\|^2 \sim N^{0.79}$. The positive-coefficient envelope follows $c_k \sim 0.54 \cdot k^{-0.24}$ for large k .

9.5 Numerical artifact in earlier computations

An earlier version of this computation (using adaptive quadrature via `scipy.quad`) reported $\alpha \approx 0.308$ and the conjecture that $\alpha = 1/3$ exactly. We have identified this as a numerical artifact: for large j, k the integrand in (5) is highly oscillatory, and adaptive quadrature without explicit cell decomposition systematically underestimates G_{jk} , inflating G_N^{-1} and causing d_N^2 to be underestimated by approximately 17% at $N = 200$. The analytic cell-summation formula (5)–(6) eliminates this error entirely. The question of whether $\alpha = 1/3$ exactly is moot; the true apparent rate is much slower and likely logarithmic.

10 Conclusion

Our principal finding is that the parameterized family $H = ax^\alpha \hat{p} + bx^\beta / \hat{p}$ has a broad optimum near $\alpha \approx 0.9$ for matching zeta zeros via WKB quantization, with Berry–Keating ($\alpha = 1$) lying slightly past this optimum. While the improvement is in the smooth part of the counting function rather than the oscillatory corrections, the computation constrains the phase space geometry of any Hilbert–Pólya candidate.

The heat kernel analysis and operator survey together constrain the underlying space to have non-integer, scale-dependent dimension ($d_{\text{eff}} \approx 2.36$ – 3.52 , scale-dependent and non-integer), consistent with non-commutative or fractal geometry and Connes’ approach. The ε_N convergence of the Connes spectral triple follows $\varepsilon_N \sim 0.606 N^{-0.226}$, reaching $\varepsilon_N < 0.1$ at $N \approx 2848$.

The Nyman–Beurling computation provides complementary evidence consistent with RH: the distance d_N decreases monotonically through $N = 5000$ with all Gram matrices full rank ($d_{5000} = 0.0723$, $\lambda_{\min} = 6.3 \times 10^{-8}$). We identify $\alpha \approx 0.308$ (reported in earlier work) as a numerical artifact of adaptive quadrature on oscillatory integrands. The corrected apparent rate ($\alpha \approx 0.071$ from $N \geq 1000$) is still decreasing and not yet asymptotic; power-law and logarithmic models are numerically indistinguishable at this scale, with both extrapolating to $d_{10^5} \approx 0.058$ – 0.061 . This underscores that the Nyman–Beurling approach cannot be made conclusive by finite computation alone, but the data are fully consistent with RH.

Our principal analytic contribution is the unconditional proof that $\lambda_n > 0$ for all $n \leq 85$ (Theorem 2), extended to all $n \leq 7.1 \times 10^6$ using the Odlyzko 2M verified-zero dataset, and further to all $n \leq 4.83 \times 10^{12}$ using the Platt–Trudgian verification of the first 3×10^{12} zeros [2] (Theorem 9). Complementing the analytic approach, the Cauchy DFT method of Section 4 provides an entirely independent numerical certificate establishing $\lambda_n > 0$ for $n \leq 10,000$ (unconditionally, no zero-location assumptions), with certified error $< 3 \times 10^{-19}$, cross-validated against published values at $n \leq 1000$ to ~ 9 – 10 significant digits, and yielding $\lambda_{10000} = 34736.5797322416$. The key structural insight is that the functional pair $F_n(\sigma, \gamma)$ is strictly positive for all $\sigma \in (0, 1)$ and all $\gamma \geq \gamma_{k+1}$ whenever the boundary condition $n \arctan(1/\gamma_{k+1}) + \arctan(\sinh(n \log r_{k+1})) < 2\pi$ holds, by a monotonicity argument on $h(\sigma) = \cosh(n \log r) \cos(n\varphi)$ exploiting $dr/d\sigma < 0$ and $d\varphi/d\sigma > 0$. This certifies positivity unconditionally on the tail zeros without assuming their location. Corollary 5 further extends the certified range through a quasi-periodic failure-window analysis, confirming $\lambda_n > 0$ at 13,881 right-endpoint values in 145 certifiable runs up to $n = 9,834,937$. The irreducible gap—verifying $\lambda_n > 0$ for *all* n —remains logically equivalent to RH and cannot be closed by finite computation.

The Lean 4 formalizations provide machine-checked infrastructure for future work.

All code, data, and results are available at <https://github.com/cascone26/riemann-hypothesis-computati>

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